Technical report - work in progress

Homotopy for rectangular multiparameter eigenvalue problems

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Abstract

A rectangular multiparameter eigenvalue problem (RMEP) is a recently introduced form of multiparameter eigenvalue problems, where we are looking for combinations of the parameters for which a multivariate rectangular matrix polynomial looses full rank. The existing numerical methods for RMEPs typically require substantial memory and are not suitable to problems with a large number of eigenvalues. We propose new numerical approach for RMEPs based on the homotopy method that has low memory requirements and, given sufficient computational time, can compute all eigenvalues. The method is also naturally parallelizable, which provides a simple and efficient way to accelerate computations. A Matlab implementation, together with a collection of test examples, is available in a public repository.

1 Introduction

We consider numerical methods for rectangular multiparameter eigenvalue problems (RMEPs) with $k \geq 2$ parameters. A generic form is

$$M(\boldsymbol{\lambda})\, x := \left(\sum_{\boldsymbol{\omega}} \boldsymbol{\lambda}^{\boldsymbol{\omega}} A_{\boldsymbol{\omega}} \right) x = 0, \tag{1}$$

where $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_k)$ is a multi-index and $A_{\boldsymbol{\omega}} = A_{\omega_1,\ldots,\omega_k}$ is an $(n+k-1) \times n$ rectangular matrix over $\mathbb{C}$. We are looking for $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_k) \in \mathbb{C}^k$ that satisfies (1) for a nonzero $x \in \mathbb{C}^n$. Such $\boldsymbol{\lambda}$ is an eigenvalue and x is the corresponding (right) eigenvector.

We will assume that M is a regular RMEP, which means that the normal rank of M, defined as

$$\operatorname{nrank}(M) := \max_{\boldsymbol{\lambda} \in \mathbb{C}^k} \operatorname{rank}(M(\boldsymbol{\lambda})),$$

is full, i.e., $\operatorname{nrank}(M) = n$. Then $\boldsymbol{\lambda} \in \mathbb{C}^k$ is an eigenvalue of (1) if and only if $\operatorname{rank}(M(\boldsymbol{\lambda})) < n$.

The degree of (1) is the maximal total degree of all monomials $\boldsymbol{\lambda}^{\boldsymbol{\omega}}$ in (1). If the degree of (1) is one, then we say that (1) is a linear RMEP. We know that a generic RMEP of degree d has

$$d^k \binom{n+k-1}{k} \tag{2}$$

eigenvalues [4].

The existing numerical methods for RMEPs are of two kind, both relying on structured matrices. The first approach is based on block Macaulay matrices [13], while the second one is based on numerical Kronecker methods for square multiparameter eigenvalue problems [4]. Although the two approaches seem to be unrelated, they both solve (1) by constructing an associated family of generalized eigenvalue problems (GEPs)

$$G_i z = \lambda_i G_0 z, \quad i = 1, \dots, k, \tag{3}$$

such that G_0 is nonsingular, matrices $G_0^{-1}G_1, \dots, G_0^{-1}G_k$ commute and $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k)$ is an eigenvalue of (1) if and only if $(\lambda_1, \dots, \lambda_k)$ is a joint eigenvalue of (3). Since the final matrices G_i in (3) are of size (2), and intermediate matrices used in the construction can be even larger, these numerical methods can in practice be applied only to problems such that (2) is not too large.

Suppose that the RMEP is so large that due to memory constraints it is not possible to explicitly compute the matrices of the associated family of GEPs (3) and then compute the eigenvalues. If we are looking for all the solutions, we need a different approach with smaller memory requirements.

A similar difficulty appears when we want to solve a (standard) square multiparameter eigenvalue problem (MEP) that has the form

$$W_i(\boldsymbol{\lambda})\, x_i := (V_{i0} + \lambda_1 V_{i1} + \cdots + \lambda_k V_{ik})\, x_i = 0, \quad i = 1, \dots, k, \tag{4}$$

where $V_{ij} \in \mathbb{C}^{n_i \times n_i}$ are square matrices, $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k) \in \mathbb{C}^k$, and $x_i \in \mathbb{C}^{n_i}$ is nonzero. In the generic case, the problem (4) has $n_1 \cdots n_k$ *eigenvalues* $\boldsymbol{\lambda} \in \mathbb{C}^k$ that are roots of the system of k multivariate characteristic polynomials equations $\det(W_i(\boldsymbol{\lambda})) = 0$ for $i = 1, \dots, k$. If $\boldsymbol{\lambda}$ and nonzero vectors $x_1, \dots, x_k$ satisfy (4), then the tensor product $x = x_1 \otimes \cdots \otimes x_k$ is called the corresponding *eigenvector*.

A MEP (4) is related to an associated system of generalized eigenvalue problems (GEPs)

$$\Delta_i z = \lambda_i\, \Delta_0 z, \qquad i = 1, \dots, k, \tag{5}$$

where the $(n_1 \cdots n_k) \times (n_1 \cdots n_k)$ matrices

$$\Delta_0 = \begin{vmatrix} V_{11} & \cdots & V_{1k} \\ \vdots & & \vdots \\ V_{k1} & \cdots & V_{kk} \end{vmatrix}_{\otimes} = \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\, V_{1\sigma_1} \otimes V_{2\sigma_2} \otimes \cdots \otimes V_{k\sigma_k},$$

where $\otimes$ denotes the Kronecker product, and

$$\Delta_i = - \begin{vmatrix} V_{11} & \cdots & V_{1,i-1} & V_{10} & V_{1,i+1} & \cdots & V_{1k} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ V_{k1} & \cdots & V_{k,i-1} & V_{k0} & V_{k,i+1} & \cdots & V_{kk} \end{vmatrix}_{\otimes}, \quad i = 1, \dots, k,$$

are called *operator determinants*, and $z = x_1 \otimes \cdots \otimes x_k$, for more details, see., e.g., [2].

If the MEP (4) is such that $n_1 \dots n_k$ is too large, then again due to memory limitations it is not possible to first explicitly construct the operator determinants Δ_i and then compute the eigenvalues from (5). An numerical method with low memory requirement that can be applied to such huge MEPs and is able to compute all solutions, is the homotopy, see, e.g., [3] and [10].

We will show that the homotopy (or numerical continuation) can be applied to solve large RMEPs as well. To the best of our knowledge, this is the first time the homotopy method has been applied to numerically solve a RMEP. In addition, we will show that the homotopy method is more efficient, sometimes even for a large margin, for many RMEPs that can be solved using methods from [4] and [13], that are implemented in [9] and [12], respectively.

2 Auxiliary results

In this section we will introduce some results on eigenvalues and eigenvectors of RMEPs that we will need later in the construction of the homotopy. The results in Theorem 2.1 and Corollary 2.2 are new and have not been published in this form before. They complement similar results for SMEPs and give additional insight into the properties of eigenvalues of RMEPs, for which the theory is much more scarce with respect to SMEPs.

2.1 The left nullspace and the left eigenvector

Let $\boldsymbol{\lambda}^{(0)}$ be an eigenvalue of the RMEP (1) that has full normal rank. The geometric multiplicity of $\boldsymbol{\lambda}^{(0)}$ is equal to the $\dim(\ker(M(\boldsymbol{\lambda}^{(0)})))$ and we say that $\boldsymbol{\lambda}^{(0)}$ is geometrically simple if $\dim(\ker(M(\boldsymbol{\lambda}^{(0)}))) = 1$.

Since matrices are rectangular and tall, there exist polynomial vectors $y(\boldsymbol{\lambda})$ such that

$$y(\boldsymbol{\lambda})^* M(\boldsymbol{\lambda}) \equiv 0.$$

The set of all such vectors forms the left nullspace $\mathcal{N}_l$ of dimension $k-1$,

$$\mathcal{N}_l(M) := \big\{ y(\boldsymbol{\lambda}) \in \mathbb{C}(\boldsymbol{\lambda})^{n+k-1} : \; y(\boldsymbol{\lambda})^* M(\boldsymbol{\lambda}) \equiv 0 \big\}.$$

There exists a polynomial basis $\{y_1(\boldsymbol{\lambda}), \ldots, y_{k-1}(\boldsymbol{\lambda})\}$ for $\mathcal{N}_l(M)$, see [5, Lemma 3.1], and then the left singular space of M at $\boldsymbol{\mu} \in \mathbb{C}^k$, which is independent on the choice of the polynomial basis, is

$$\ker_{\boldsymbol{\mu}}(M) := \operatorname{span}(y_1(\boldsymbol{\mu}), \ldots, y_{k-1}(\boldsymbol{\mu})).$$

If $\boldsymbol{\lambda}^{(0)}$ is an eigenvalue, then $\dim(\ker(M(\boldsymbol{\lambda}^{(0)})^*)) \geq k$ and each vector $y \notin \ker_{\boldsymbol{\lambda}^{(0)}}(M)$ such that $y^* M(\boldsymbol{\lambda}^{(0)}) = 0$ is a representative of a left eigenvector. If $\lambda^{(0)}$ is geometrically simple, then $\dim(\ker(M(\boldsymbol{\lambda}^{(0)})^*)) = k$ and there exist k linearly independent vectors $z_1, \ldots, z_k$ such that $z_i^* M(\boldsymbol{\lambda}^{(0)}) = 0$ for $i = 1, \ldots, k$.

2.2 Nonsingularity of the Jacobian matrix

Let $f_{j_1,\ldots,j_n}(\lambda) := \det(M_{j_1,\ldots,j_n}(\boldsymbol{\lambda}))$ be an $n \times n$ minor of $M(\boldsymbol{\lambda})$ composed from rows $j_1, \ldots, j_n$, where $1 \leq j_1 < \ldots < j_n \leq n+k-1$, note that there are $\binom{n+k-1}{n}$ minors of this kind. Then $\boldsymbol{\lambda}^{(0)}$ is an eigenvalue of M, see, e.g., [5], if and only if

$$f_{j_1,\ldots,j_n}(\boldsymbol{\lambda}^{(0)}) = 0 \quad \text{for all} \quad 1 \leq j_1 < \ldots < j_n \leq n+k-1. \tag{6}$$

If $\boldsymbol{\lambda}^{(0)}$ is an eigenvalue of M, then it is algebraically simple if it is a simple common root of (6), which is true if and only if the Jacobian of (6) has full rank at $\boldsymbol{\lambda}^{(0)}$, where the Jacobian is a $\binom{n+k-1}{n} \times k$ matrix with rows

$$\left[\frac{\partial f_{j_1,\ldots,j_n}}{\partial \lambda_1}(\boldsymbol{\lambda}) \quad \cdots \quad \frac{\partial f_{j_1,\ldots,j_n}}{\partial \lambda_k}(\boldsymbol{\lambda}) \right], \quad 1 \leq j_1 < \ldots < j_n \leq n+k-1. \tag{7}$$

Each algebraically simple eigenvalue is also geometrically simple, while the opposite is not always true. The following result gives a condition, which is much easier to check than the rank of the Jacobian (7), that a geometrically simple eigenvalue $\boldsymbol{\lambda}^{(0)}$ must satisfy to be algebraically simple.

Theorem 2.1. *Let $\boldsymbol{\lambda}^{(0)}$ be an algebraically simple eigenvalue of a RMEP $M(\boldsymbol{\lambda})$. If x is the right eigenvector and $z_1, \ldots, z_k$ are linearly independent vectors from $\ker(M(\boldsymbol{\lambda}^{(0)})^*)$, then the matrix*

$$\begin{bmatrix} z_1^* \frac{\partial M}{\partial \lambda_1}(\boldsymbol{\lambda}^{(0)})x & \cdots & z_1^* \frac{\partial M}{\partial \lambda_k}(\boldsymbol{\lambda}^{(0)})x \\ \vdots & & \vdots \\ z_k^* \frac{\partial M}{\partial \lambda_1}(\boldsymbol{\lambda}^{(0)})x & \cdots & z_k^* \frac{\partial M}{\partial \lambda_k}(\boldsymbol{\lambda}^{(0)})x \end{bmatrix} \tag{8}$$

is nonsingular.

Proof. Let $P_{j_1,\ldots,j_n}$, where $1 \le j_1 < \ldots < j_n \le n+k-1$, be an $(n+k-1) \times n$ matrix

$$P_{j_1,\ldots,j_n} = \begin{bmatrix} e_{j_1} & e_{j_2} & \cdots & e_{j_n} \end{bmatrix}$$

whose columns are standard basis vectors. Then

$$P_{j_1,\ldots,j_n}^T M(\boldsymbol{\lambda}) = M_{j_1,\ldots,j_n}(\boldsymbol{\lambda})$$

and $\det(P_{j_1,\ldots,j_n}^T M(\boldsymbol{\lambda})) = f_{j_1,\ldots,j_n}(\boldsymbol{\lambda})$.

Since $\boldsymbol{\lambda}^{(0)}$ is algebraically simple, the rank of the Jacobian (7) is full, therefore we can select k matrices $P_1, \ldots, P_k$ of the above form that correspond to k linearly independent rows of (7) at $\boldsymbol{\lambda} = \boldsymbol{\lambda}^{(0)}$. When we multiply (1) by $P_1^T, \ldots, P_k^T$, we obtain a polynomial MEP

$$P_i^T M(\boldsymbol{\lambda}) = \left(\sum_{\boldsymbol{\omega}} \boldsymbol{\lambda}^{\boldsymbol{\omega}} P_i^T A_{\boldsymbol{\omega}} \right) x = 0, \quad i = 1, \ldots, k, \tag{9}$$

such that $\boldsymbol{\lambda}^{(0)} = (\lambda_1, \ldots, \lambda_k)$ is an eigenvalue of (9) and $x \otimes \cdots \otimes x$ is the corresponding eigenvector. Note that the eigenvalues of the MEP (9) are roots of the system of polynomial equations

$$\det(P_i^T M(\boldsymbol{\lambda})) = 0, \quad i = 1, \ldots, k, \tag{10}$$

which is a subset of equations (6), therefore the MEP (9) has additional eigenvalues that are not eigenvalues of the RMEP M. Since we have selected $P_1, \ldots, P_k$ from k linearly independent rows of (7) at $\boldsymbol{\lambda} = \boldsymbol{\lambda}^{(0)}$, the Jacobian of (10) is nonsingular at $\boldsymbol{\lambda} = \boldsymbol{\lambda}^{(0)}$ and it follows that $\boldsymbol{\lambda}^{(0)}$ is an algebraically simple eigenvalue of the MEP (9).

Let $y_1 \otimes \cdots \otimes y_k$ be the left eigenvector of $\boldsymbol{\lambda}^{(0)}$ for the MEP (9). Since $\boldsymbol{\lambda}^{(0)}$ is algebraically simple, it follows that the matrix

$$\begin{bmatrix} y_1^* P_1^T \frac{\partial M}{\partial \lambda_1}(\boldsymbol{\lambda}^{(0)})x & \cdots & y_1^* P_1^T \frac{\partial M}{\partial \lambda_k}(\boldsymbol{\lambda}^{(0)})x \\ \vdots & & \vdots \\ y_k^* P_k^T \frac{\partial M}{\partial \lambda_1}(\boldsymbol{\lambda}^{(0)})x & \cdots & y_k^* P_k^T \frac{\partial M}{\partial \lambda_k}(\boldsymbol{\lambda}^{(0)})x \end{bmatrix} \tag{11}$$

is nonsingular. This follows from the result [8, Propositon 3.2] for $k = 2$ that can be generalized to an arbitrary number of parameters in a straightforward way.

Since $y_i^* P_i^T M(\boldsymbol{\lambda}^{(0)}) = 0$ for $i = 1, \ldots, k$, vectors $z_1 = P_1 y_1, \ldots, z_k = P_k y_k$ belong to $\ker(M(\boldsymbol{\lambda}^{(0)})^*)$. Clearly, the matrix (8) is singular if vectors $z_1, \ldots, z_k$ are linearly dependent, therefore, as the matrices (11) and (8) are equal and nonsingular, vectors $z_1, \ldots, z_k$ have to be linearly independent. This proofs the theorem for a particular choice of basis $z_1, \ldots, z_k$ for $\ker(M(\boldsymbol{\lambda}^{(0)})^*)$ and it is easy to see that the theorem then holds for any basis $z_1, \ldots, z_k$ of $\ker(M(\boldsymbol{\lambda}^{(0)})^*)$. □

This is a generalization of the well-known result for matrix polynomials which says that if y and x are the left and the right eigenvectors of a polynomial eigenvalue problem $P(\lambda)x = 0$ for a simple eigenvalue λ_0, then $y^*P'(\lambda_0)x \neq 0$, see, e.g., [6, Theorem 4.5].

The above result can be written in a simpler way for a linear RMEP.

Corollary 2.2. *Let $\boldsymbol{\lambda}^{(0)}$ be an algebraically simple eigenvalue of a linear RMEP*

$$M(\boldsymbol{\lambda})x = (A_0 + \lambda_1 A_1 + \ldots + \lambda_k A_k)x = 0. \tag{12}$$

If x is the right eigenvector and $z_1, \ldots, z_k$ are linearly independent vectors from $\ker(M(\boldsymbol{\lambda}^{(0)})^*)$, then the matrix

$$\begin{bmatrix} z_1^* A_1 x & \cdots & z_1^* A_k x \\ \vdots & & \vdots \\ z_k^* A_1 x & \cdots & z_k^* A_k x \end{bmatrix} \tag{13}$$

is nonsingular.

3 Homotopy for linear RMEPs

Let us first show that all numerical methods for MEPs could be applied to RMEPs as well, thus this applies also to the existing homotopy methods for MEPs such as [3] and [10]. Suppose that we have a linear RMEP

$$(A_0 + \lambda_1 A_1 + \ldots + \lambda_k A_k)x = 0, \tag{14}$$

where $A_i \in \mathbb{C}^{(n+k-1)\times n}$ for $i = 0, \ldots, k$. As shown in [4, Theorem 1], if we multiply (14) by random matrices $P_1, \ldots, P_k$ of size $n \times (n+k-1)$, we obtain a MEP

$$(P_i A_0 + \lambda_1 P_i A_1 + \ldots + \lambda_k P_i A_k)x_i = 0, \quad i = 1, \ldots, k, \tag{15}$$

such that if $(\lambda_1, \ldots, \lambda_k)$ is an eigenvalue of (14) and $x \neq 0$ is the corresponding eigenvector, then $(\lambda_1, \ldots, \lambda_k)$ is an eigenvalue of (15) and $x \otimes \cdots \otimes x$ is the eigenvector. So, we can compute all eigenvalues of (15) using a homotopy method for MEPs and then extract the eigenvalues that are also solutions of (14). The problem with this approach is that the problem (15) has n^k eigenvalues that include $\binom{n+k-1}{k}$ eigenvalues of (14). If n^k is much larger then $\binom{n+k-1}{k}$ then this approach is not efficient.

If we apply a homotopy method for MEPs to the above transformation, then we first have to follow all n^k paths to obtain all eigenvalues of the MEP and then in the second phase we can extract the eigenvalues of (14) by computing the residuals of the RMEP. The difference can be huge, for instance, for $n = 7$ and $k = 6$ the problem (15) has 117649 eigenvalues but only 924 of them are also eigenvalues of (14). Clearly, it would be much more efficient if we could follow only 924 paths to get the solutions of (14). To do this, we need a homotopy method tailored specifically to RMEPs.

The common idea of all homotopy methods, see, e.g., [1], is the following. Suppose that we want to solve a nonlinear equation $f(z) = 0$ that has m solutions. We construct the initial nonlinear equation $g(z) = 0$ of the same type with the same number of solutions in such way that we know its solutions $\alpha_1, \ldots, \alpha_m$. Now, we construct a homotopy $h(t, z)$ that depends smoothly on the parameter $t \in [0, 1]$ so that $h(0, z) = g(z)$, $h(1, z) = f(z)$, and $h(t, z) = 0$ has m solutions for all $t \in [0, 1]$, e.g., $h(t, z) := (1 - t)g(t) + tf(t)$. Assuming that there are m disjoint paths $\alpha_i(t)$ such that $h(t, \alpha_i(t)) = 0$ for $t \in [0, 1]$, $\alpha_i(0) = \alpha_i$, we follow these m paths from $t = 0$ to $t = 1$ to get the solution of $f(z) = 0$.

To apply the above idea to the linear RMEP (14), we need to be able to construct an initial RMEP of the same size and number of parameters that we can solve efficiently.

3.1 Construction of an initial RMEP

We will construct the initial RMEP as

$$(B_0 + \lambda_1 B_1 + \ldots + \lambda_k B_k)x = 0, \tag{16}$$

where B_0 is a random banded matrix with nonzero entries on the main diagonal and the next $k-1$ subdiagonals, i.e., $(B_0)_{ij} = 0$ for $i < j$ or $i - j > k$, and

$$B_i = \begin{bmatrix} 0_{(i-1)\times n} \\ I_n \\ 0_{(k-i)\times n} \end{bmatrix}, \quad i = 1, \ldots, k, \tag{17}$$

are shifted identity matrices. RMEPs of the above form are generic, see [11], therefore we can expect that (16), where nonzero entries of B_0 are selected as random complex numbers from the normal distribution, has $\binom{n+k-1}{k}$ simple eigenvalues.

3.1.1 Case $k = 2$

For $k = 2$ we can simply write down the eigenvalues and eigenvectors of (16). Let us denote the elements of the i-th diagonal of B_0 as $a_1^{(i)}, \ldots, a_k^{(i)}$. We get

$$B_0 + \lambda_1 B_1 + \lambda_2 B_2 = \begin{bmatrix} a_1^{(0)} + \lambda_1 & & & \\ a_1^{(1)} + \lambda_2 & a_2^{(0)} + \lambda_1 & & \\ & \ddots & \ddots & \\ & & a_{n-1}^{(1)} + \lambda_2 & a_n^{(0)} + \lambda_1 \\ & & & a_n^{(1)} + \lambda_2 \end{bmatrix}.$$

The rank of the above matrix is smaller than n if we set to zero one diagonal and one subdiagonal element from different rows. It is easy to see that the eigenvalues of (16) are

$$(\lambda_1, \lambda_2) = (-a_j^{(0)}, -a_\ell^{(1)}), \quad 1 \le j \le \ell \le n,$$

which gives $\binom{n+1}{2}$ eigenvalues that we are looking for. Clearly, all eigenvalues are simple for generic values $a_i^{(j)}$, $i = 1, \ldots, n$, $j = 0, 1$.

3.1.2 Case $k = 3$

For $k = 3$ we have to solve several one-parameter eigenvalue problems to obtain all solutions. Now

$$B_0 + \lambda_1 B_1 + \lambda_2 B_2 + \lambda_3 B_3 = \begin{bmatrix} a_1^{(0)} + \lambda_1 & & & & \\ a_1^{(1)} + \lambda_2 & a_2^{(0)} + \lambda_1 & & & \\ a_1^{(2)} + \lambda_3 & a_2^{(1)} + \lambda_2 & a_3^{(0)} + \lambda_1 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-2}^{(2)} + \lambda_3 & a_{n-1}^{(1)} + \lambda_2 & a_n^{(0)} + \lambda_1 \\ & & & a_{n-1}^{(2)} + \lambda_3 & a_n^{(1)} + \lambda_2 \\ & & & & a_n^{(2)} + \lambda_3 \end{bmatrix}. \tag{18}$$

Similarly as for $k = 2$, we set $\lambda_1 = -a_j^{(0)}$ and $\lambda_3 = -a_\ell^{(2)}$ for $1 \le j < \ell \le k$ to zero out one element on the upper diagonal and one on the lower diagonal of (18) from different rows. The rank of (18) is then smaller than n if the submatrix of (18) with rows $j+1, \ldots, \ell$ and columns $j, \ldots, \ell$ is singular, i.e., when

$$\left(\begin{bmatrix} a_j^{(1)} & a_{j+1}^{(0)} - a_j^{(0)} & & \\ a_j^{(2)} - a_\ell^{(2)} & \ddots & \ddots & \\ & \ddots & \ddots & a_\ell^{(0)} - a_j^{(0)} \\ & & a_{\ell-1}^{(2)} - a_\ell^{(2)} & a_\ell^{(1)} \end{bmatrix} + \lambda_2 I \right) w = 0 \tag{19}$$

for a nonzero w. By solving the square eigenvalue problem (19) we get $\ell - j$ values of λ_2 for each pair $\lambda_1 = -a_j^{(0)}$ and $\lambda_3 = -a_\ell^{(2)}$ for $1 \le j < \ell \le k$. This gives all $\binom{n+2}{3}$ eigenvalues of the initial problem.

3.1.3 Case $k \ge 4$

To find the solutions of the initial RMEP in case $k \ge 4$ we apply a similar approach as in the previous case, with a difference that now we have to solve a smaller $(k-2)$-parameter linear RMEP to get values $\lambda_2, \ldots, \lambda_{k-1}$ for each pair of $\lambda_1 = -a_j^{(0)}$, $\lambda_k = -a_\ell^{(k-1)}$ for $1 \le j < j+k-1 \le \ell \le k$ from the submatrix of (16) composed of rows $j+1, \ldots, \ell+k-2$ and columns $j, \ldots, \ell$. We provide the construction in Algorithm 1.

Algorithm 1: Construct initial linear RMEP.

Input: n: number of columns, k: number of parameters.
Output: Matrices $B_0, \ldots, B_k$ and eigenpairs $\big((\lambda_1^{(j)}, \ldots, \lambda_k^{(j)}), x_j\big)$ for $j = 1, \ldots, \binom{n+k-1}{k}$.

1: Select random entries $a_i^{(j)}$, $i = 1, \ldots, n$, $j = 0, \ldots, k-1$, of B_0.
2: for $j = 1, \ldots, n$
3: for $\ell = j+k-1, \ldots, n$
4: $\widetilde{B}_i = B_i(j+1 : \ell+k-2, j : \ell)$ for $i = 0, \ldots, k$
5: Compute eigenpairs $\big((\mu_2^{(p)}, \ldots, \mu_{k-1}^{(p)}), z_p\big)$, $p = 1, \ldots, r = \binom{\ell-j+k-2}{k-2}$, of the RMEP $\big(\widetilde{B}_0 - a_j^{(0)}\widetilde{B}_1 - a_\ell^{(k-1)}B_k + \sum_{i=2}^{k-1} \mu_i \widetilde{B}_i\big)z = 0$
6: Add $\big((-a_j^{(0)}, \mu_2^{(p)}, \ldots, \mu_{k-1}^{(p)}, -a_\ell^{(k-1)}), [0_{j-1}\ z^T\ 0_{n+k-1-\ell}]^T\big)$, $p = 1, \ldots, r$, to the list.

The recursive construction in Algorithm 1 requires numerical solutions of smaller RMEPs with $k-2$ parameters. We will see that in most cases the most efficient approach to solve these subproblems is to recursively use the final homotopy algorithm.

For many combinations of n and k it is possible to efficiently construct the initial RMEP. This becomes even more important for polynomial RMEPs where we can obtain an initial polynomial RMEP from a linear RMEP with a simple transformation.

3.2 Path following

For the homotopy we will apply the homogeneous version of the RMEP

$$(\lambda_0 A_0 + \lambda_1 A_1 + \ldots + \lambda_k A_k)x = 0, \tag{20}$$

where we additionally require that $(\lambda_0, \lambda_1, \ldots, \lambda_k) \neq (0, \ldots, 0)$. In the homogeneous setting, eigenvalues of (20) with $\lambda_0 = 0$ correspond to infinite eigenvalues of (14) and each finite eigen-

value $(\lambda_1, \ldots, \lambda_k)$ of (14) corresponds to the eigenvalue $(1, \lambda_1, \ldots, \lambda_k)$ of (20). In a similar way we transform (16) into a homogeneous RMEP

$$(\lambda_0 B_0 + \lambda_1 B_1 + \ldots + \lambda_k B_k)x = 0, \tag{21}$$

and construct the homotopy as

$$H(t, x, \boldsymbol{\lambda}) = \begin{bmatrix} \sum_{i=0}^{k} \lambda_i \left((1-t)B_i + tA_i\right) x \\ x^* x - 1 \\ \boldsymbol{\lambda}^* \boldsymbol{\lambda} - 1 \end{bmatrix}, \tag{22}$$

where $\boldsymbol{\lambda} = (\lambda_0, \lambda_1, \ldots, \lambda_k)$. The bottom two rows in (22) normalize the eigenvector x and the homogeneous eigenvalue $\boldsymbol{\lambda}$ that both have to be nonzero. Clearly, the first row of $H(0, x, \boldsymbol{\lambda})$ is $\lambda_0 B_0 + \ldots + \lambda_k B_k$ and the first row of $H(1, x, \boldsymbol{\lambda})$ is $\lambda_0 A_0 + \ldots + \lambda_k A_k$, therefore, as t moves from 0 to 1, the initial RMEP (21) smoothly changes to the RMEP (22) that we want to solve.

Let $(\boldsymbol{\lambda}^{(0)}, x^{(0)})$ be an eigenpair of (21) such that $\|\boldsymbol{\lambda}^{(0)}\|_2 = 1$ and $\|x^{(0)}\|_2 = 1$. We assume that there exists a smooth path $(x(t), \boldsymbol{\lambda}(t))$ such that $x(0) = x^{(0)}$, $\boldsymbol{\lambda}(0) = \boldsymbol{\lambda}^{(0)}$, and $H(t, x(t), \boldsymbol{\lambda}(t)) = 0$ for $0 \le t \le 1$. Starting at $(\boldsymbol{\lambda}^{(0)}, x^{(0)})$ we numerically follow this path from $t = 0$ to $t = 1$ using the standard predictor-corrector scheme, where we first use Euler's method to obtain an approximation for the next point on the path and then correct it with Newton's method.

Suppose that $H(t, x, \boldsymbol{\lambda}) = 0$ and h is the stepsize. To predict the next point on the path at $t + h$, we use Euler's predictor. By differentiating $H(t, x, \boldsymbol{\lambda}) = 0$ we get the linear system for the tangent vector

$$J(t, x, \boldsymbol{\lambda}) \begin{bmatrix} \dot{x} \\ \dot{\lambda}_0 \\ \vdots \\ \dot{\lambda}_k \end{bmatrix} = \begin{bmatrix} \sum_{i=0}^{k} \lambda_i (B_i - A_i) x \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \tag{23}$$

where

$$J(t, x, \boldsymbol{\lambda}) = \begin{bmatrix} \sum_{i=0}^{k} \lambda_i \left((1-t)B_i + tA_i\right) & ((1-t)B_0 + tA_0)x & \cdots & ((1-t)B_k + tA_k)x \\ 2x^* & 0 & \cdots & 0 \\ 0 & 2\overline{\lambda_0} & \cdots & 2\overline{\lambda_k} \end{bmatrix} \tag{24}$$

is the corresponding Jacobian matrix. Note that the last two rows of (22) are not complex differentiable. However, it is known in the context of optimization problems that we can employ the approximation (24), for the details, see [7].

We solve (23), move in the direction of the tangent, and take

$$x_{\mathrm{P}} = x + h\dot{x}, \quad \boldsymbol{\lambda}_{\mathrm{P}} = \boldsymbol{\lambda} + h \begin{bmatrix} \dot{\lambda}_0 & \cdots & \dot{\lambda}_k \end{bmatrix}^T$$

for the predictor. Then we use Newton's method with the initial approximation $(x_{\mathrm{P}}, \boldsymbol{\lambda}_{\mathrm{P}})$ to compute the solution of (22) at $t + h$.

In one step of Newton's method we update $(x_r, \boldsymbol{\lambda}_r)$ to $(x_{r+1}, \boldsymbol{\lambda}_{r+1})$ as

$$x_{r+1} = x_r + \Delta x, \quad \boldsymbol{\lambda}_{r+1} = \boldsymbol{\lambda}_r + [\Delta\lambda_0 \ \cdots \ \Delta\lambda_k]^T,$$

where the correction is the solution of the linear system

$$J(t+h, x_r, \boldsymbol{\lambda}_r) \begin{bmatrix} \Delta x \\ \Delta\lambda_0 \\ \vdots \\ \Delta\lambda_k \end{bmatrix} = -H(t+h, x_r, \boldsymbol{\lambda}_r). \tag{25}$$

Let $C_i(t) = (1-t)B_i(t) + tA_i(t)$ for $i = 0, \ldots, k$ and $G(t, \boldsymbol{\lambda}) = \sum_{i=0}^{k} \lambda_i C_i(t)$. Then we can write the homotopy (22) as

$$H(t, x, \boldsymbol{\lambda}) = \begin{bmatrix} G(t, \boldsymbol{\lambda})x \\ x^*x - 1 \\ \boldsymbol{\lambda}^*\boldsymbol{\lambda} - 1 \end{bmatrix}.$$

Theorem 3.1. *If $\boldsymbol{\lambda}$ is a simple eigenvalue of $G(t, \boldsymbol{\lambda})$ for $t \in [0, 1)$, and x is the corresponding eigenvector, then the Jacobian matrix $J(t, x, \boldsymbol{\lambda})$ is nonsingular.*

Proof. Suppose that

$$J(t, x, \lambda) \begin{bmatrix} z \\ \boldsymbol{\mu} \end{bmatrix} = 0, \tag{26}$$

where $z \in \mathbb{C}^n$ and $\boldsymbol{\mu} = [\mu_0 \ \cdots \ \mu_k]^T$. Then

$$G(t, \lambda)z + \mu_0 C_0(t)x + \ldots + \mu_k C_k(t)x = 0. \tag{27}$$

Let $y_1, \ldots, y_k$ be linearly independent vectors from $\ker(G(t, \boldsymbol{\lambda})^*)$. If we multiply (27) by $y_1, \ldots, y_k$, we get

$$\begin{bmatrix} y_1^* C_0(t)x & \cdots & y_1^* C_k(t)x \\ \vdots & & \vdots \\ y_k^* C_0(t)x & \cdots & y_k^* C_k(t)x \end{bmatrix} \begin{bmatrix} \mu_0 \\ \vdots \\ \mu_k \end{bmatrix} = 0. \tag{28}$$

It follows from a homogeneous version of Theorem 2.2 that the $k \times (k+1)$ matrix in (28) has full rank, so its nullspace has dimension 1, therefore $\boldsymbol{\mu}$ and $\boldsymbol{\lambda}$ are colinear and $\boldsymbol{\mu} = \alpha\boldsymbol{\lambda}$ for a scalar α. But then it follows from $2\boldsymbol{\lambda}^*\boldsymbol{\mu} = 0$ that $\alpha = 0$ and $\boldsymbol{\mu} = 0$. From (27) it now follows that $z = \beta x$ for a scalar β, because $G(t, \boldsymbol{\lambda})$ also has nullspace of dimension 1. Then it follows from $2x^*z = 0$ that $\beta = 0$. So, since the only solution for (26) is $z = 0$ and $\boldsymbol{\mu} = 0$, this means that $J(t, x, \boldsymbol{\lambda})$ is nonsingular. □

It follows from the above theorem that the singular points of the homotopy are exactly the points where $\boldsymbol{\lambda}$ is a multiple eigenvalue of $G(t, \boldsymbol{\lambda})$. There can be only finitely many such points on $t \in [0, 1)$, moreover, we expect that the initial problem is generic enough so that the probability of hitting a singular point with a homotopy path is zero. If the final RMEP has a multiple eigenvalue of multiplicity r, then r homotopy paths will meet at $t = 1$.

When we follow the paths numerically, it can happen that we jump from one path to another. This can happen if e.g., the stepsize h is too large or the Jacobian matrix is close to singular. A simple solution is to retrace all paths that lead to a cluster of $k > 1$ eigenvalues such that the corresponding eigenvectors are not linearly independent, using a smaller stepsize h and other settings that make the computation more accurate. Since more accurate computations also require more time, it makes sense to use these settings only for the paths that have to be retraced.

4 Homotopy for polynomial RMEPs

Compared to a linear RMEP with the same number of parameters k and columns n, a generic polynomial RMEP of degree d has d^k times as many eigenvalues. Even if the number of eigenvalues $\binom{n+k-1}{k}$ of the linear RMEP is such that the problem of that size can in practice be solved with the existing methods from [13] and [4], when we multiply this by d^k the problem quickly becomes too large and with too many solutions for the mentioned methods.

Suppose that for n and k we can construct an initial linear RMEP

$$(B_0 + \mu_1 B_1 + \ldots + \mu_k B_k)x = 0, \tag{29}$$

with Algorithm 1 or, if this is more efficient, compute the eigenpairs of the initial linear RMEP with methods from [13] and [4]. If we apply the substitutions $\mu_i = \lambda_i^k$ for $i = 1, \ldots, k$, we obtain an initial polynomial RMEP of degree d

$$(B_0 + \lambda_1^d B_1 + \ldots + \lambda_k^d B_k)x = 0. \tag{30}$$

Clearly, if $(\mu_1, \ldots, \mu_k)$ is an eigenvalue of (29), then $(\mu_1^{1/d}, \ldots, \mu_k^{1/d})$ is an eigenvalue of (30). Since we get d different roots for each component, we get d^k different eigenvalues of (30) from each eigenvalue of (29). They share the same eigenvector x which is the same for (29) and (30).

So, as long as we can construct and solve the initial linear RMEP, we can apply the homotopy method to solve the polynomial RMEP. In many situations, when $\binom{n+k-1}{k}$ is not too large but $d^k\binom{n+k-1}{k}$ is, this gives a numerical approach that can compute all the solutions, and to the best of our knowledge, this is at this moment the only numerical method capable of this. The path following part may be slow as we can have many paths to follow, but it does not require a lot of memory, so the only limitation is time. We can improve the later by following the paths in parallel. Note that the method is so embarrassingly simple to implement in parallel, that we can expect a speedup of factor M using M processors.

Suppose that we want to solve a polynomial RMEP of degree d that we write in the homogeneous form as

$$P(\boldsymbol{\lambda})x := \left(\sum_{|\boldsymbol{\omega}|=d} \boldsymbol{\lambda}^{\boldsymbol{\omega}} A_{\boldsymbol{\omega}} \right) x = 0, \tag{31}$$

where $\boldsymbol{\lambda} = (\lambda_0, \lambda_1, \ldots, \lambda_k)$, $\boldsymbol{\omega} = (\omega_0, \ldots, \omega_k)$, and $|\boldsymbol{\omega}| = \omega_0 + \cdots + \omega_k$. The homogeneous version of (30) is

$$Q(\boldsymbol{\lambda})x := \left(\sum_{i=0}^{k} \lambda_i^d B_i \right) x = 0, \tag{32}$$

note that the initial RMEP contains only univariate monomials. Now we define $R(t, \boldsymbol{\lambda}) = (1-t)Q(\boldsymbol{\lambda}) + tP(\lambda)$ and use the homotopy

$$H(t, x, \boldsymbol{\lambda}) = \begin{bmatrix} R(t, \boldsymbol{\lambda})x \\ x^* x - 1 \\ \boldsymbol{\lambda}^* \boldsymbol{\lambda} - 1 \end{bmatrix}. \tag{33}$$

The corresponding Jacobian is

$$J(t, x, \boldsymbol{\lambda}) = \begin{bmatrix} R(t, \boldsymbol{\lambda}) & \dfrac{\partial R}{\partial \lambda_0}(t, \boldsymbol{\lambda})x & \cdots & \dfrac{\partial R}{\partial \lambda_k}(t, \boldsymbol{\lambda})x \\ 2x^* & 0 & \cdots & 0 \\ 0 & 2\overline{\lambda_0} & \cdots & 2\overline{\lambda_k} \end{bmatrix}. \tag{34}$$

and for the tangent vector we have to solve

$$J(t,x,\boldsymbol{\lambda})\begin{bmatrix}\dot{x}\\ \dot{\lambda}_0\\ \vdots\\ \dot{\lambda}_k\end{bmatrix}=\begin{bmatrix}Q(\boldsymbol{\lambda})x-P(\boldsymbol{\lambda})x\\ 0\\ \vdots\\ 0\end{bmatrix}. \tag{35}$$

We use Euler's predictor as for the linear RMEP and then apply Newton's method to compute the next point on the path. In each step we have to solve the linear system (25) with (34) and (33).

4.1 Special cases of polynomial RMEPs

A k-parameter polynomial RMEP of degree d with matrices of size $(n+k-1)\times n$ can have less than (2) eigenvalues if not all monomials of degree d are present. For instance, while a generic 3-parameter RMEP of degree 2 has $8\binom{n+2}{3}$ eigenvalues, a generic RMEP of the form

$$(A_{000}+\lambda_1 A_{100}+\lambda_2 A_{010}+\lambda_3 A_{001}+\lambda_1^2 A_{200})x=0 \tag{36}$$

has only $2\binom{n+2}{3}$ solutions. We can exploit this and reduce the number of paths to follow. For the particular case (36) we can use an initial RMEP of the form

$$(B_0+\lambda_1^2 B_1+\lambda_2 B_2+\lambda_3 B_k)x=0 \tag{37}$$

that has the same number of solutions. We construct (37) from a linear RMEP (29) in a similar way as in the generic case with all monomials, by applying the substitutions $\mu_1=\lambda_1^2$, $\mu_2=\lambda_2$, and $\mu_3=\lambda_3$.

5 Numerical examples

All numerical experiments were carried out in Matlab 2023a on a desktop PC with 64 GB RAM and i7–11700K 3.6 GHz CPU. For following the paths in parallel 8 workers were used. In the recursive construction of the initial RMEP methods from MultiParEig [9] and MacaulayLab [12] are used to solve smaller RMEPs when estimated that this is more efficient than using the homotopy. The Matlab implementation of the homotopy method together with all numerical examples is available in the public repository `https://github.com/borplestenjak/RMEP_homotopy`.

Example 5.1. We test the homotopy method on random linear RMEPs with matrices, whose elements are real standard normal variables. Such generic problems are regular and such that all eigenvalues are simple and finite. We compared the timings and accuracy to methods `rect_multipareig` from [9] and `macaulaylab` from [12].

Example 5.2. In a similar way we test the homotopy method on random quadratic RMEPs with matrices, whose elements are real standard normal variables. Such generic problems are regular and such that all eigenvalues are simple and finite. We compared the timings and accuracy to methods `rect_quad_multipareig` from [9] and `macaulaylab` from [12].

Example 5.3. In a similar way we test the homotopy method on random polynomial RMEPs of degree three or above, with matrices, whose elements are real standard normal variables. Such generic problems are regular and such that all eigenvalues are simple and finite. We compared the timings and accuracy to method `macaulaylab` from [12].

Example 5.4. The homotopy methods can be applied to problems with infinite eigenvalues as well.

5.1 RMEPs with positive dimensional solutions

Although we do not provide any supporting theory, the homotopy method works in principle also for RMEPs with positive dimensional solutions. As a proof of concept we give several numerical examples. What happens for such a problem if the Jacobian is nonsingular for all $t \in [0, 1)$ and no path jumping occur, is that some of the homotopy paths diverge to infinity, some converge to points on a positive dimensional solution, and the remaining paths converge to the eigenvalues.

The main problem with such examples is that it might be very difficult or even impossible to detect the paths that have to be retraced in case of path jumping. Namely, if a path that in the exact computation converges to a finite eigenvalue, numerically converges to a point on a positive dimensional solution, we will not be able to detect it. If we know the exact number of finite eigenvalues and we find out that the number of computed finite eigenvalues is less than that, we know that some paths did not converge to right solutions. But since these paths lead to different points on a positive dimensional solution, they all appear simple. So, since we can not detect individual bad paths, we have to retrace all paths that converged to a positive dimensional solution and there can be many. But, because the retracing is much slower, we want to apply it only to few paths, otherwise the whole method becomes inefficient.

Example 5.5. Simple example with positive definite solution.

Example 5.6. LTI example.

Example 5.7. ARMA example.

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